

# BENJAMIN BLACK

Postdoctoral researcher with a professional background in conservation, specialising in socio-environmental systems modelling to support evidence-based policy and planning. Passionate about advancing open science, communicating research beyond academia, and mentoring others in modelling, programmatic data analysis, and visualisation. Experienced in managing small and large teams, especially in cross-cultural and interdisciplinary settings, with a strong commitment to collaborative and inclusive research.

## EDUCATION

- Doctor of Civil Engineering**  
ETH Zürich, chair of Planning of Landscape Urban Systems (PLUS)  
Zürich, Switzerland  
Thesis: *Identifying effective area-based conservation strategies under uncertainties*  
Awards: *Nominated for the ETH medal for outstanding doctoral theses*
- MSc in Environmental Sciences, Policy and Management (MESPOM)**  
Central European University, University of Manchester, Lund University, University of the Aegean  
Hungary, Greece, UK  
Thesis: *Assessing the ecological effectiveness of protected areas in Cambodia: a quasi-experimental counterfactual estimation of avoided deforestation*  
Awards: *Outstanding academic achievement (highest GPA in cohort: 4.00/4.00)*
- BSc with 1st Class Honours in Biological Sciences**  
Edinburgh Napier University  
Edinburgh, Scotland  
Thesis: *Simulating the future potential of degraded Mangrove plots using field data from Kenya*  
Awards: *University medal (highest average mark on degree route); Lawrence Ho student prize; Dennis Aw award.*

## PROFESSIONAL EXPERIENCE

- Freelance Research Consultant**  
Prognos AG  
Zürich, Switzerland  
Performed Land Use and Land Cover change modelling of Shared Socioeconomic Pathways for Switzerland as part of project funded by the Swiss National Centre for Climate Services
- Biodiversity and SMART Intern**  
The Wildlife Conservation Society (WCS)  
Phnom Penh, Cambodia  
Developed a programmatic routine to expedite the collection of external data from the IUCN, EDGE ranking and CITES to produce protected area species inventories in line with Darwin Core standards. Assisted in the training of new community biodiversity rangers in the process of SMART data collection in the field.
- Student Affairs Manager**  
The School for Field Studies Center for Conservation and Development Studies in the Lower Mekong  
Siem Reap, Cambodia  
Coordinated safety and risk management as well as logistics for all centre activities including multi day field trips to remote locations. Served as the medical point person, facilitating first aid training for staff, while also supporting student well-being and cultural adaptation in a pastoral role.



## CONTACT INFO

Post-Doctoral researcher  
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<https://www.researchgate.net/profile/Benjamin-Black-5>

## TECHNICAL SKILLS

Advanced: R (Shiny, Quarto), Dinamica EGO, Netica and MS Office  
Intermediate: ArcGIS, QGIS, CSS and Git  
Basic: Python, HTML, Docker, JavaScript, Bash and Slurm

## LANGUAGES

English: Native  
German: B2

- April. 2015  
|  
June. 2017

● **Project Manager & Volunteer Co-ordinator**  
Elephant Livelihood Initiatives and Environment (E.L.I.E)  
📍 Mondulkiri, Cambodia

*Managed a diverse team (Bunong Indigenous minority, and Khmer) of 48 staff. Oversaw the operation of an ecotourism project receiving 4000+ visitors a year. Facilitated ongoing capacity building of staff and development of the volunteer program. Compiled budgets, managed cash flow, and created detailed financial and organizational reports.*
- Oct. 2015  
|  
Dec. 2015

● **Independent Consultant Srae Y Community ecotourism project**  
World Wide Fund for Nature (WWF)  
📍 Mondulkiri, Cambodia

*Undertook site visits to assess the status of the project and prepared a report, detailing strategies to make the project operational and sustainable. Planned and delivered a series of training sessions to project staff and stakeholders from other communities engaged in ecotourism.*
- May 2013  
|  
Apr. 2015

● **Volunteer Development Officer & Conservation Support Assistant**  
The Conservation Volunteers (TCV)  
📍 Central Scotland

*Planned, developed and delivered the Ochils Landscape Partnerships volunteering program and two TCV community health projects. Engaged with, and ensured the safety and wellbeing of a diverse cross section of volunteers, many with physical or mental disabilities as well as young adults/children.*



## TEACHING, TRAINING & WORKSHOP DELIVERY

- Feb. 2021  
|  
Jul. 2024

● **Participatory Environmental Modelling: MSc course at ETH Zürich**  
*Designed and taught an MSc-level course integrating students and practitioners to develop simulation models for real-world environmental decision-making.*  
📍 Zürich, Switzerland
- Sep. 2024

● **Reproducible Research with R and Quarto: Workflows for data, projects and publications**  
*Three hour workshop as part of Landscape 2024 conference*  
📍 Berlin, Germany

 Website



## GRANTS & FUNDING

- Dec. 2024

● **Future development of biodiversity and ecosystem services as a basis for the revision of the sectoral plan for biodiversity**  
Canton of Bern Department of Agriculture and Nature  
📍 Bern, Switzerland  
*Project in collaboration with the Wyss Academy for Nature to generate policy-relevant evidence supporting the canton's revision of the sectoral biodiversity plan.*
- Jun. 2023

● **Nature-positive Scenarios for Environmental Transitions: Peru (NASCENT-PERU)**  
Swiss Re Foundation  
📍 Zürich, Switzerland  
*Funded project under the SwissRE Biodiversity and Ecosystem Services Scenarios Modelling Initiative*

 Website



## PUBLICATIONS

- In Review

● **Identifying robust conservation strategies to secure ecosystem service provision under uncertainties**  
Black, B., Adde, A., Külling, N., Büth, C., Kurmann, M., Lehmann, A., Altermatt, F., Guisan, A., Gret-Regamey, A.  
*Global Environmental Change*
- In Review

● **Temporal Archetypes of Nature's Contributions to People for Sustainable Landscape Development**  
Wicki, S., Black, B., Külling, N., Kurmann, M., Wang, J., Lehmann, A., Gret-Regamey, A.  
*Regional Environmental Change*

In Review

● **Where will the change come from? Identifying decision-relevant factors in future Land Use and Cover Change ({LUCC}) scenarios to support planning under uncertainty**

Roman, O., Adey, B., [Black, B.](#), Zeindl, E.

*Futures*

In Review

● **The PAVE pathway: how to talk about models with people who did(n't) build them**

Bell, A., Troost, C., Filatova, T., Vincenot, C., Qin, C., Fiedler, S., Lemmen, C., Lynch, I., Pierce, M., Edmonds, B., McIntire, E., Wang, H., Koralewski, T., Grant, W., Wang, M., [Black, B.](#), Grimm, V.

*Bioscience*

2024

● **Broadening the horizon in land use change modelling: Normative scenarios for nature positive futures in Switzerland**

[Black, B.](#), Adde, A., Farinotti, D., Guisan, A., Külling, N., Kurmann, M., Martin, C., Mayer, P., Rabe, S., Streit, J., Zekollari, H., Grêt-Regamey, A.

*Regional Environmental Change*

 Read  Data  Code

2023

● **Archetypes of social-ecological-technological systems for managing ecological infrastructure**

Wicki, S., [Black, B.](#), Kurmann, M., Grêt-Regamey, A.

*Environmental Research Letters*

 Read

2023

● **Re-considering the status quo: Improving calibration of land use change models through validation of transition potential predictions**

[Black, B.](#), van Strien, M. J., Adde, A., Grêt-Regamey, A.

*Environmental Modelling & Software*

 Read  Code

2022

● **Counterfactual assessment of protected area avoided deforestation in Cambodia: Trends in effectiveness, spillover effects and the influence of establishment date**

[Black, B.](#), Anthony, B. P.

*Global Ecology and Conservation*

 Read  Data



## PRESENTATIONS

Jun. 2025

● **A “Scope, Simulation and Story” approach to identify future spatial development scenarios**

Computational Urban Planning and Urban Management Conference 2025

 University College London

 Slides

Sep. 2024

● **Negotiating scenarios of a desirable ecological infrastructure for Switzerland**

4th Swiss Landscape Congress

 Tramelan, Switzerland

 Slides

Jun. 2024

● **Normative scenarios of landscape change for Nature-Positive futures in Peru**

World Biodiversity Forum

 Davos, Switzerland

 Slides

Nov. 2023

● **Broadening the scope of future visions for nature positive futures in Peru**

Swiss Re Institute Resilience Summit 2023

 Zürich, Switzerland

 Slides

Sep. 2022

● **Characterising non-stationarity in predictive models of land use in Swiss mountain parks to inform scenarios for deliberative transformation**

International Mountain conference 2022

 Innsbruck, Austria

 Slides

Jul. 2022



**ValPar.CH: Integrating land use change, Ecosystem Service and Biodiversity modelling to simulate pathways for a functioning Ecological Infrastructure for Switzerland**

11th International Environmental Modelling and Software Society conference



Brussels, Belgium

 Slides

Jul. 2022



**Combining filter and embedded approaches to improve variable selection in land use change Cellular Automata models**

11th International Environmental Modelling and Software Society conference



Brussels, Belgium

 Slides

Jun. 2019



**Ex-ante identification of policy interventions to secure a functioning ecological infrastructure: A participatory Bayesian Network approach in Switzerland**

3rd Ecosystem Services Partnership Europe Regional conference



Tartu, Estonia

 Slides